

Towards Real-Time Precision Agriculture: A Task-Aware Multi-Modal Decision Framework

Sam Guessoum
Worcester Polytechnic Institute
Worcester, MA, USA
mssguessoum@wpi.edu

Naima Boukhiar
University of Algiers 1
Algiers, Algeria
naima.boukhiar@univ-alger.dz

Abstract—Precision agriculture requires real-time, data-driven decision-making to optimize resource use under environmental constraints. However, existing solutions often address tasks such as irrigation, crop recommendation, and disease detection independently, leading to fragmented systems.

This paper presents a task-aware multi-modal framework integrating IoT sensing, cloud processing, and adaptive machine learning into a unified decision-support pipeline. The system processes both structured sensor data and image data, dynamically selecting models based on modality, latency, and computational cost.

Experimental results show strong performance, achieving 99.80% accuracy for irrigation prediction, 93.93% for crop recommendation, and 83.50% for disease detection, with end-to-end latency between 3 and 6 seconds. A preliminary proof-of-concept deployment indicated an approximate 22% reduction in fertilizer usage and approximately 99.6% system uptime; these figures are early, informally observed indicators rather than outcomes of a controlled, systematically documented field trial.

These results demonstrate an efficient and deployable approach for integrated precision agriculture, with rigorous multi-site longitudinal field validation identified as the next step.

Index Terms—Precision Agriculture, Intelligent Systems, IoT, Machine Learning, Adaptive Systems, Multi-Modal Learning, Smart Farming

I. INTRODUCTION

Agriculture is undergoing a critical transformation driven by increasing global food demand, climate variability, and resource scarcity. Global food production must increase by approximately 70% by 2050, while agriculture already accounts for nearly 70% of global freshwater withdrawals. These constraints necessitate intelligent, data-driven approaches to improve efficiency and sustainability.

Traditional agricultural practices rely on uniform field management and periodic manual sampling, failing to capture spatial and temporal variability and leading to inefficient use of water, fertilizers, and pesticides, with studies reporting up to 30–50% nutrient loss due to leaching and runoff.

Precision agriculture addresses these challenges through IoT-enabled sensing and machine learning [2], [3], [15]. However, most existing systems focus on isolated tasks such as irrigation or disease detection, resulting in fragmented solutions that lack integrated intelligence.

To address these limitations, this paper proposes a task-aware multi-modal framework that integrates IoT sensing, cloud-based analytics, and machine learning into a unified

real-time decision-support framework. The system processes heterogeneous data sources, including soil nutrient measurements (N, P, K), environmental conditions, and plant images, to support multiple tasks such as irrigation prediction, crop recommendation, fertilizer optimization, and plant disease detection.

A key contribution of this work is the introduction of a task-aware model selection strategy that dynamically allocates computational resources based on input modality and task complexity. Lightweight models are used for structured sensor data to ensure low-latency inference, while deep learning models are applied to image-based tasks requiring higher representational capacity. This design enables the system to balance predictive accuracy and computational efficiency in real-time deployment scenarios. The main contributions of this work are summarized as follows:

- A unified adaptive multi-modal intelligent system for real-time precision agriculture integrating IoT, cloud computing, and machine learning.
- A task-aware model selection strategy enabling efficient allocation of computational resources across heterogeneous data modalities.
- A comprehensive evaluation framework including structured datasets, image-based tasks, and real-world deployment validation.
- Demonstration of practical impact through improved resource efficiency and system reliability in field conditions.

The remainder of this paper is organized as follows. Section II presents the system architecture of the proposed intelligent framework. Section III describes the methodology, including the task-aware model selection strategy and learning models. Section IV introduces the proof-of-concept prototype and its implementation. Section V presents the experimental results and performance analysis. Finally, Section VI concludes the paper and outlines directions for future work.

II. SYSTEM ARCHITECTURE

The proposed system is designed as a multi-modal intelligent architecture that enables real-time sensing, data processing, and decision-making for precision agriculture. The architecture integrates IoT sensing, edge-level preprocessing,

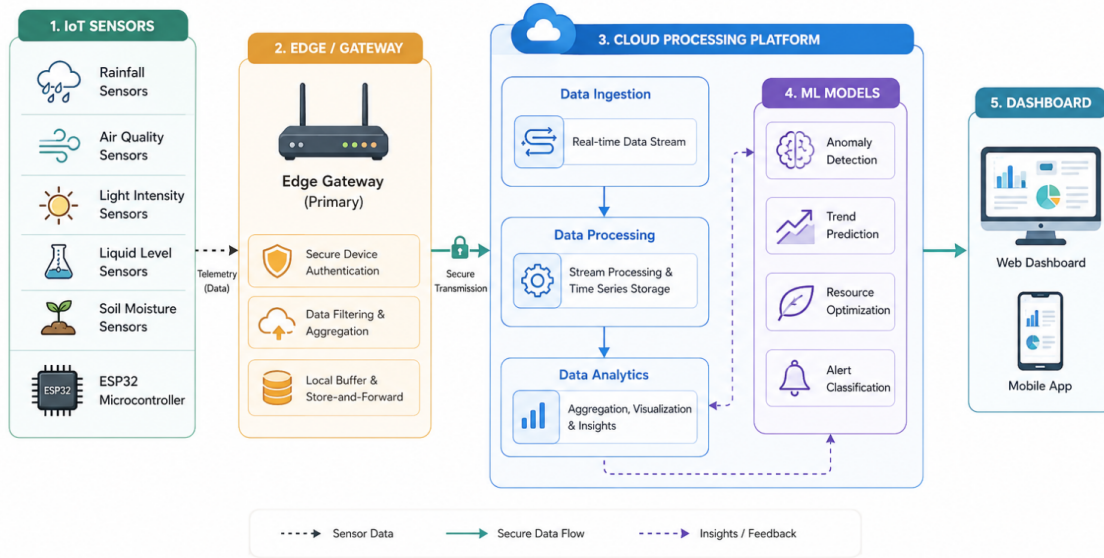


Fig. 1. Simplified end-to-end architecture of the proposed adaptive multi-modal intelligent system, showing the flow from IoT sensing and edge preprocessing to cloud analytics, machine learning inference, and decision-support visualization.

cloud-based analytics, machine learning inference, and user-facing decision support into a unified pipeline. Such IoT-driven architectures have been shown to significantly enhance real-time monitoring and resource optimization in agricultural systems [10].

A. System Overview

The system architecture (Fig. 1) follows an end-to-end pipeline that transforms sensor data into actionable recommendations. IoT sensors collect environmental and soil data, which is preprocessed at the edge gateway through filtering and buffering. The processed data is transmitted to the cloud, where it is stored, analyzed, and used for machine learning inference. The resulting predictions are delivered through a dashboard and mobile interface, enabling real-time monitoring and decision support.

B. Sensing and Edge Processing

The sensing layer captures multi-modal environmental data through a set of distributed IoT sensors. These sensors measure key agricultural variables such as soil moisture, rainfall, air quality, and light intensity, which are critical for decision-making processes such as irrigation and crop recommendation.

The ESP32 microcontroller acts as the central interface between sensors and the rest of the system. It supports multiple communication protocols and periodically polls sensor readings. Before transmission, data undergoes preprocessing at the edge gateway, including noise filtering, validation, and aggregation.

The edge gateway also implements security mechanisms such as device authentication to ensure trusted data transmission, and a local buffering mechanism allows temporary storage of sensor readings in case of network interruptions. This layer reduces data redundancy and ensures that only

consistent, validated data is forwarded to the cloud, improving overall system efficiency.

C. Cloud Processing and Decision Support

The cloud processing platform is responsible for data ingestion, storage, analytics, and machine learning inference. Incoming sensor data is streamed in real time and processed through a sequence of stages, including aggregation, time-series storage, and feature extraction.

Machine learning models are integrated within the cloud layer to perform task-specific predictive analytics. For structured sensor data, Gaussian Naive Bayes is used for crop and fertilizer recommendation due to its computational efficiency and probabilistic formulation, while a Decision Tree model is used for irrigation prediction to capture threshold-based relationships between environmental variables. For image-based plant disease detection, a ResNet-based convolutional neural network is employed to extract hierarchical visual features and classify disease patterns. These models are selected using the task-aware strategy described in Section III, ensuring a balance between accuracy, latency, and computational cost. Structured data supports irrigation, crop, and fertilizer tasks, while image data is used for disease detection.

The outputs of these models are combined to generate actionable recommendations. These recommendations are delivered to the user through a web dashboard and mobile application, enabling real-time monitoring and informed decision-making. The integration of sensing, processing, and intelligent analytics within a unified architecture allows the system to operate as a real-time decision-support framework for precision agriculture.

Table I summarizes the layered architecture, from sensing and edge preprocessing through cloud analytics to the dashboard and mobile application layer. This separation of

TABLE I
SYSTEM LAYER OVERVIEW

Layer	Components	Function
Sensing	IoT sensors, ESP32	Data acquisition
Edge Processing	Gateway, filtering, buffering	Preprocessing and validation
Cloud Processing	Data ingestion, storage, ML models	Analytics and inference
Application	Dashboard, mobile app	Visualization and decision support

concerns improves system scalability, reliability, and real-time responsiveness.

III. METHODOLOGY

The proposed system is formulated as an adaptive multi-modal intelligent framework designed to support real-time decision-making in precision agriculture. The methodology integrates multi-task learning, task-aware model selection, and real-time data processing into a unified pipeline. This approach ensures that different agricultural tasks are handled using models that are best suited to their data characteristics and computational requirements.

A. Multi-Task Problem Formulation

The system is modeled as a multi-task learning problem:

$$T = \{T_{crop}, T_{irrigation}, T_{fertilizer}, T_{disease}\} \quad (1)$$

Each task in T corresponds to a specific agricultural decision process: T_{crop} (crop recommendation), $T_{irrigation}$ (irrigation prediction), $T_{fertilizer}$ (fertilizer recommendation), and $T_{disease}$ (plant disease detection).

Each task is defined as a mapping:

$$f_i(x_i) \rightarrow y_i \quad (2)$$

where x_i is the input feature representation for task T_i , f_i is the predictive model, and y_i is the output prediction. For example, in irrigation prediction, x_i includes environmental variables such as soil moisture and temperature, while y_i is a binary irrigation decision. In disease detection, x_i is an image input, and y_i is the predicted disease class.

The system processes two categories of input data. Structured sensor data is represented as:

$$x_t = [N, P, K, M, T_s, H_s, pH, R, T_a, H_a] \quad (3)$$

Here, N , P , and K represent soil macronutrients; M represents soil moisture; T_s and H_s represent soil temperature and humidity; pH represents soil acidity; R represents rainfall; and T_a , H_a represent atmospheric conditions. These variables directly influence agricultural decisions such as irrigation scheduling and crop selection.

For plant disease detection, the input is an image tensor:

$$I \in \mathbb{R}^{H \times W \times C} \quad (4)$$

where H and W denote image dimensions and C represents the number of channels. This representation enables extraction of visual features such as color variation, texture, and lesion patterns.

B. Task-Aware Model Selection

A key contribution of this work is the task-aware model selection strategy, which assigns models based on task complexity, input modality, and computational constraints.

The optimal model is selected according to:

$$M^* = \arg \min_M (\alpha E(M) + \beta L(M) + \gamma C(M)) \quad (5)$$

In this formulation, $E(M)$ represents prediction error, $L(M)$ represents inference latency, and $C(M)$ represents computational cost, with α , β , and γ controlling the accuracy–speed–efficiency trade-off. These coefficients were set qualitatively rather than tuned via formal multi-objective optimization, weighting latency and cost more heavily for structured-data tasks and accuracy more heavily for the image-based task; formalizing this weighting is left as future work. For crop recommendation, this trade-off was evaluated empirically: as detailed in Section V-A, Random Forest reached higher accuracy (99.09% vs. 93.93%) but required roughly $100\times$ longer training time (1.74 s vs. 0.017 s), motivating GNB under the latency-weighted objective. For irrigation and disease detection, model choice instead followed established suitability for the data modality (threshold-based splits for sensor-driven irrigation decisions; residual CNNs for hierarchical visual features [8]) rather than an equally exhaustive benchmark; extending systematic multi-model comparison to these tasks is future work. Gaussian Naive Bayes is additionally used for fertilizer recommendation, for the same efficiency rationale.

The prediction rule for Gaussian Naive Bayes is:

$$P(y|x) \propto P(y) \prod_k P(x_k|y) \quad (6)$$

This equation estimates the posterior probability of class y given input features x . $P(y)$ represents the prior probability of a class, while $P(x_k|y)$ represents the likelihood of observing feature x_k given class y . The independence assumption allows efficient computation, making it suitable for real-time applications.

For irrigation prediction, the Decision Tree uses information gain:

$$IG(S, A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v) \quad (7)$$

Here, $H(S)$ is the entropy of the dataset before splitting, representing uncertainty. S_v represents subsets after splitting on feature A , and the weighted sum of entropies measures the remaining uncertainty. Maximizing information gain selects the feature that best reduces uncertainty, enabling interpretable decision rules.

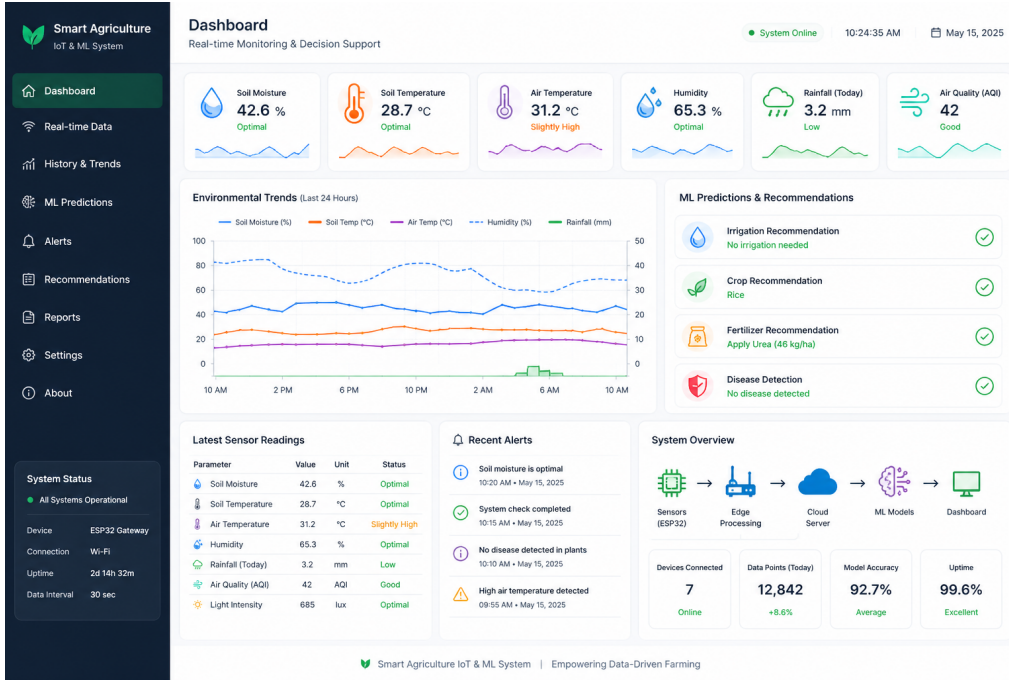


Fig. 2. Web application interface displaying real-time sensor data and system-generated recommendations.

For disease detection, the ResNet model uses residual learning:

$$H(x) = F(x) + x \quad (8)$$

In this formulation, $F(x)$ represents the learned transformation and x is the identity shortcut connection. This structure allows the network to learn residual mappings, improving gradient flow and enabling deeper architectures to capture complex visual patterns.

C. Training Protocol and Decision Generation

Before training, structured data is standardized:

$$x' = \frac{x - \mu}{\sigma} \quad (9)$$

where μ and σ are the mean and standard deviation. This normalization ensures that all features contribute equally during training.

The model training objective is defined as:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \ell(f(x_i), y_i) \quad (10)$$

where ℓ is the loss function. For classification tasks, cross-entropy loss is used, as it penalizes incorrect predictions proportionally to their confidence.

The datasets are partitioned to ensure robust evaluation. The crop recommendation dataset (2200 samples), obtained from a publicly available Kaggle dataset, includes soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH,

and rainfall, and is widely used for crop prediction tasks in precision agriculture [11].

The irrigation dataset (100,002 samples) consists of structured environmental and soil-related variables such as soil moisture, temperature, and rainfall, enabling binary irrigation decision modeling. The dataset is derived from publicly available smart irrigation and environmental monitoring datasets commonly used in precision agriculture research [14], and is split into 80% training, 10% validation, and 10% testing.

The plant disease dataset (approximately 87,000 images across 38 classes) is based on the widely used PlantVillage dataset, which contains labeled images of healthy and diseased plant leaves across multiple crop species [12], and is split into 70% training, 15% validation, and 15% testing.

Due to its small size (99 samples), the fertilizer dataset, sourced from a publicly available agricultural dataset, includes soil nutrient composition (N, P, K), crop type, and environmental conditions for fertilizer recommendation tasks, and is evaluated using 5-fold cross-validation to reduce overfitting [13].

The final system output is generated through decision fusion:

$$R = \Phi(y_{crop}, y_{irrigation}, y_{fertilizer}, y_{disease}) \quad (11)$$

Here, R represents the final recommendation vector, and Φ is the decision fusion function that integrates predictions from all tasks. This function ensures that outputs are consistent and actionable, transforming model predictions into practical agricultural decisions such as irrigation scheduling, fertilizer adjustment, and crop selection.

IV. PROOF-OF-CONCEPT PROTOTYPE

To validate the usability of the proposed system, a proof-of-concept dashboard prototype was developed as the user-facing interface for real-time agricultural monitoring and decision support. The prototype connects the sensing and cloud layers to an interactive visualization environment, allowing users to interpret field conditions and model outputs in a practical format.

As shown in Fig. 2, the dashboard presents live environmental readings, including soil moisture, soil temperature, air temperature, humidity, rainfall, air quality, and light intensity. These values are displayed through summary cards and a latest-readings table, enabling users to quickly assess current field conditions. The interface also includes trend visualizations over time, which support historical analysis and help identify changes in soil and environmental behavior. The dashboard further integrates decision-support outputs generated by the machine learning models. These include irrigation recommendations, crop recommendations, fertilizer guidance, and disease detection alerts. By presenting model predictions alongside raw sensor readings, the prototype allows users to connect environmental conditions with actionable agricultural decisions. The prototype validates the end-to-end workflow by transforming sensor data and model outputs into an interpretable interface for real-time agricultural decision-making.

V. RESULTS

This section presents a detailed evaluation of the proposed adaptive multi-modal intelligent system across all agricultural tasks. The analysis focuses on predictive performance, feature contribution, and system-level efficiency, demonstrating the effectiveness of the task-aware model selection strategy.

A. Predictive Performance

The quantitative results summarized in Table II demonstrate the effectiveness of the proposed system across all tasks. Precision and recall were additionally computed for irrigation prediction (below); for crop recommendation, fertilizer recommendation, and disease detection, the original evaluation protocol recorded accuracy and macro-averaged F1-score, with full per-class precision/recall identified as near-term follow-up work to enable finer-grained error analysis, particularly given class imbalance in the crop and disease datasets.

TABLE II
MODEL PERFORMANCE SUMMARY

Task	Acc.	F1	Size	Model
Irrigation	99.80%	0.998	100K	Decision Tree
Crop Type	93.93%	0.938	2,200	GNB
Fertilizer	$\approx 100\%$	≈ 1.00	99	GNB
Disease	83.50%	0.835	87K	ResNet

The irrigation prediction task, formulated as a binary classification problem, achieved an accuracy of 99.80% using a Decision Tree model. The evaluation was conducted on a dataset of 100,002 samples, with an 80%/10%/10% train-validation-test split. The model also achieved a precision of 0.9981, recall of 0.9980, and an F1-score of 0.9980 on the test set. The high performance indicates that irrigation decisions are strongly correlated with environmental variables such as soil moisture and rainfall.

For the crop recommendation task, Gaussian Naive Bayes achieved an accuracy of 93.93% on a dataset of 2200 samples, using a 70%/15%/15% split. The model achieved a macro-averaged F1-score of 0.938, demonstrating balanced performance across multiple crop classes. While more complex models such as Random Forest achieved higher accuracy (up to 99.09%), Gaussian Naive Bayes was selected due to its significantly lower training time (0.017 seconds compared to 1.74 seconds), making it more suitable for real-time deployment, as detailed in Section III-B.

For fertilizer recommendation, the dataset consisted of only 99 samples, and 5-fold cross-validation was applied. Gaussian Naive Bayes achieved an average accuracy of 100% across folds, with negligible variance. Given the very limited sample size, this near-perfect result should be interpreted cautiously and does not by itself establish generalization; evaluating this task on a substantially larger, independently collected dataset is a priority for future work, and the fertilizer output should currently be treated as advisory.

The plant disease detection task was evaluated on approximately 87,000 images across 38 classes, using a 70%/15%/15% split. The ResNet-based model achieved a validation accuracy of 83.50% and a training accuracy of approximately 88.20%, indicating a moderate generalization gap. The macro F1-score on the validation set was approximately 0.835, reflecting the increased difficulty of image-based classification compared to structured data tasks.

B. Feature Contribution Analysis

To quantify the importance of individual input features, an ablation study was conducted. The contribution of each feature is defined as:

$$\Delta Acc_k = Acc_{full} - Acc_{without_k} \quad (12)$$

where Acc_{full} represents the accuracy using all features and $Acc_{without_k}$ represents the accuracy after removing feature k .

The results indicate that soil macronutrients (N, P, K) are the most influential features for crop and fertilizer recommendation, with an average accuracy decrease of approximately 6.5% when removed. For irrigation prediction, soil moisture was identified as the dominant feature, resulting in a decrease of approximately 4.2% when excluded. Environmental variables such as temperature and humidity contributed consistently across tasks, with performance reductions between 3.1% and 4.8%.

These results confirm that combining multiple sensor modalities improves predictive performance by capturing complementary environmental factors.

C. System Efficiency and Latency

The system was evaluated in terms of end-to-end latency, defined as:

$$L_{total} = L_{edge} + L_{transmission} + L_{cloud} + L_{inference} \quad (13)$$

Empirical measurements indicate that edge processing latency ranges from 0.5 to 1.2 seconds, transmission latency ranges from 1.0 to 3.0 seconds depending on network conditions, cloud processing latency ranges from 0.8 to 1.5 seconds, and model inference latency ranges from 0.01 seconds for lightweight models to approximately 0.5 seconds for the ResNet model.

The total system latency therefore ranges between 3 and 6 seconds, enabling near real-time decision-making. The use of lightweight models for structured tasks significantly reduces computational overhead, allowing the system to maintain responsiveness despite the inclusion of deep learning components.

A preliminary proof-of-concept deployment additionally suggested an approximate 22% reduction in fertilizer usage and approximately 99.6% uptime; however, site details, deployment duration, and a controlled baseline were not systematically logged, so these are early directional indicators rather than validated field-trial results, and a controlled multi-site longitudinal trial is planned to confirm them.

Related integrated systems [16]–[18] typically target one or two tasks (97–99.95% acc.) and do not report end-to-end latency; this framework unifies four tasks with reported latency (3–6 s), though datasets and protocols differ across systems and are not directly comparable.

Remaining limitations include unablated preprocessing (Eq. (9) standardization and categorical encoding), untested scalability and fault tolerance beyond edge buffering (Section II-B) under many concurrent devices, and security limited to device authentication without encryption; addressing these, alongside formalizing α, β, γ , is future work.

VI. CONCLUSION

This paper presented a task-aware multi-modal framework for real-time precision agriculture, integrating IoT sensing, cloud-based analytics, and machine learning into a unified decision-support system. Unlike conventional approaches that treat agricultural tasks independently, the proposed framework enables coordinated decision-making across irrigation prediction, crop recommendation, fertilizer optimization, and plant disease detection.

A key contribution is the task-aware model selection strategy, which dynamically assigns models based on input modality, computational cost, and latency constraints. This design allows the system to leverage lightweight models for structured sensor data while applying deep learning models to complex

image-based tasks, achieving an effective balance between predictive accuracy and computational efficiency.

Experimental results demonstrate strong performance across all tasks, including 99.80% accuracy for irrigation prediction, 93.93% for crop recommendation, and 83.50% validation accuracy for disease detection, with end-to-end latency between 3 and 6 seconds. A preliminary deployment additionally suggested fertilizer savings and high uptime, though, as discussed in Section V, these remain early indicators pending controlled multi-site validation.

Overall, the proposed framework demonstrates that combining multi-modal sensing with adaptive, task-aware intelligence can improve efficiency and decision-making in precision agriculture, with the limitations identified in Section V defining a clear path toward a more rigorously validated deployment.

REFERENCES

- [1] A. Kamilaris and F. X. Prenafeta-Boldú, “Deep learning in agriculture: A survey,” *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018.
- [2] M. Ayaz, M. Ammad-Uddin, I. Baig, A. Aggoune, and M. Al-Husseini, “Internet-of-Things (IoT)-based smart agriculture: Toward making the fields talk,” *IEEE Access*, vol. 7, pp. 129551–129583, 2019.
- [3] L. García, L. Parra, J. M. Jimenez, J. Lloret, and P. Lorenz, “IoT-based smart irrigation systems: An overview on the recent trends on sensors and IoT systems for irrigation in precision agriculture,” *Sensors*, vol. 20, no. 4, 2020.
- [4] K. P. Ferentinos, “Deep learning models for plant disease detection and diagnosis,” *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.
- [5] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [6] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*. Springer, 2009.
- [7] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *arXiv preprint arXiv:1412.6980*, 2014.
- [8] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE CVPR*, 2016, pp. 770–778.
- [9] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
- [10] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, “Internet of Things (IoT): A vision, architectural elements, and future directions,” *Future Generation Computer Systems*, vol. 29, no. 7, pp. 1645–1660, 2013.
- [11] Kaggle, “Crop Recommendation Dataset,” Available: <https://www.kaggle.com/datasets/arkabhowmik/crop-recommendation>
- [12] D. Hughes and M. Salathé, “An open access repository of images on plant health,” *arXiv:1511.08060*, 2015.
- [13] Kaggle, “Fertilizer Recommendation Dataset,” Available: <https://www.kaggle.com/datasets/miadul/fertilizer-recommendation-dataset>
- [14] UCI Machine Learning Repository, “Smart Irrigation Dataset,” Available: <https://archive.ics.uci.edu/>
- [15] M. Javaid, A. Haleem, R. P. Singh, and R. Suman, “Understanding the adoption of Industry 4.0 technologies in agriculture: Applications and challenges,” *Sustainable Operations and Computers*, vol. 3, pp. 305–318, 2022.
- [16] S. Douzi, M. Bouni, B. Hssina, and K. Douzi, “Integrated IoT approaches for crop recommendation and yield-prediction using machine-learning,” *IoT*, vol. 5, no. 4, pp. 634–649, 2024.
- [17] C. Musanase, A. Vodacek, D. Hanyurwimfura, A. Uwitonze, and I. Kabandana, “Data-driven analysis and machine learning-based crop and fertilizer recommendation system for revolutionizing farming practices,” *Agriculture*, vol. 13, no. 11, p. 2141, 2023.
- [18] N. P. Sable, V. K. Shukla, P. N. Mahalle, and V. Khedkar, “Hybrid multimodal learning framework for crop disease detection, adaptive treatment, and price forecasting,” *Frontiers in Computer Science*, vol. 8, art. 1855851, 2026.